Pyramids of Energy

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 17 February 2025

SaveMyExams

Did this video help you? Yes No

Pyramids of Energy: Extended

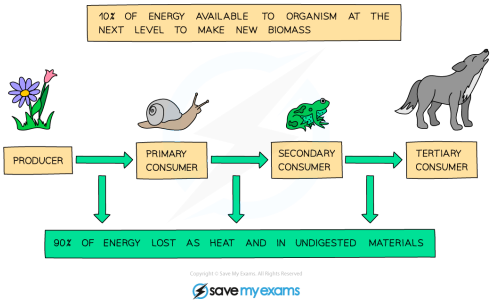
Extended Tier Only

- The transfer of stored energy in food chains is not always efficient, and **energy is lost to the environment between each trophic level**; reasons for this inefficiency include:
  - not all of the stored energy consumed by an organism goes into making new cells that can be eaten; energy may instead be transferred to:
    - **excretion of metabolic waste**, e.g. urine that gets removed from the organism
    - **movement**
    - **heat**
  - few organisms eat or digest an entire organism, so any energy stored in the uneaten or undigested parts does not get passed on, e.g.
    - energy stored in **uneaten** parts, such as roots of plants, or bones of animals does not enter the consumer's body at all
    - **undigested waste** (faeces) is removed from the body and provides food for decomposers
- Only **around 10 %** of the energy stored at each trophic level is available to the next trophic level

Help me understand

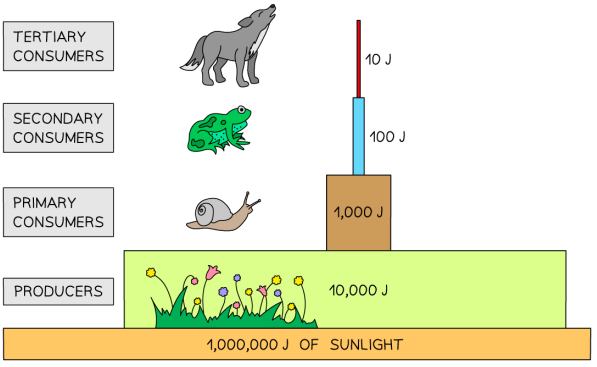
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- In order to survive a consumer would have to eat a huge number of prey organisms every day to get the amount of energy needed; this is unlikely to be possible



Energy is lost at each trophic level

- The energy stored at each trophic level can be represented visually in a **pyramid of energy**



- Pyramids of energy can have advantages over pyramids of biomass because:
  - they highlight the **energy lost** at each trophic level
  - different organisms may not all store the same amount of energy in their biomass, so pyramids of energy allow for **more accurate comparisons**

Energy transfer in a human food chain

- Humans are **omnivores**, obtaining energy from both plants and animals, and this gives us a **choice of what we eat**
- These choices, however, have an **impact on what we grow** and how we use ecosystems
- Think of the following food chains, both involving humans:

wheat → cow → human

wheat → human

- Given what we know about **energy transfer in food chains**, it is clear that if humans eat the wheat there is **much more energy available** to them than if they eat the cows that eat the wheat
  - This is because **energy is lost from the cows**, so there is less available to pass on to humans

cannot eat (e.g. grass) or that are too widely distributed for us to collect (e.g. algae in the ocean which form the food of fish we eat), so the situation is more complex than for the example shown above

Examiner Tips and Tricks

Make sure you read the question carefully and tailor your answer to the specific organism you are being asked about – e.g. plants do not produce urine or faeces so you could not give this as one of the ways in which they transfer energy to the environment!

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Solution Banks
Sitemap
Settings
Log out

Company
About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects
Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.